

Scattering by a Central Potential

3.1 Kinematics and the Differential Cross Section

$$\text{Incident energy: } E = \frac{m}{2} v_0^2, \quad \text{Incident velocity: } v_0 = \sqrt{2E/m}, \quad (3.1)$$

$$\text{Angular momentum: } \ell = s m v_0 = s \sqrt{2mE}, \quad s = \frac{\ell}{\sqrt{2mE}}. \quad (3.2)$$

The volume element in spherical coordinates is

$$d^3r = r^2 dr \underbrace{d\varphi \sin \theta}_{d\Omega} d\theta, \quad (3.3)$$

and since the azimuthal angle φ is irrelevant by symmetry, integrating it out gives the solid angle subtended by a ring of scattering angles between θ and $\theta + d\theta$:

$$d\Omega = 2\pi \sin \theta d\theta. \quad (3.4)$$

Because the impact parameter and the scattering angle are in one-to-one correspondence, $s \leftrightarrow \theta$, the deflection is fully controlled by the function $s(\theta, E)$ (Fig. 3.1).

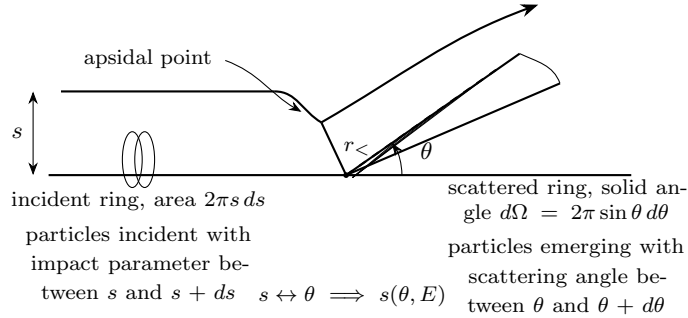


Figure 3.1: Particles incident in the ring of area $2\pi s ds$ emerge into the ring of scattering angles between θ and $\theta + d\theta$; $r_<$ is the distance of closest approach, reached at the apsidal point. The azimuthal angle is irrelevant by symmetry.

3.1.1 Incident Intensity and the Differential Cross Section

$$I = \frac{\# \text{ particles incident per unit time}}{\text{incident beam cross-section area}}, \quad (3.5)$$

$$\sigma(\Omega) d\Omega = \frac{\# \text{ particles scattered into } d\Omega \text{ per unit time}}{\text{incident intensity}}. \quad (3.6)$$

$$\begin{aligned}
I(2\pi s)|ds| &= \# \text{ particles going through the ring between } s \text{ and } s + ds \\
&= \# \text{ particles emerging with angle between } \theta \text{ and } \theta + d\theta \\
&= I \sigma(\theta) 2\pi \sin \theta |d\theta|.
\end{aligned} \tag{3.7}$$

Therefore

$$\boxed{\sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right|} \tag{3.8}$$

and the total cross section is defined by

$$\boxed{\sigma_T = \int d\Omega \sigma(\Omega)} . \tag{3.9}$$

3.2 General Relation between Impact Parameter and Scattering Angle



Figure 3.2: The apsidal angle ψ and the scattering angle θ in the repulsive and attractive cases; r_m is the distance of closest approach.

REPULSIVE CASE	ATTRACTIVE CASE
$\pi = \theta + 2\psi$	$\pi = 2\psi - \theta$
$\theta = \pi - 2\psi$	$\theta = 2\psi - \pi$
$\psi = \frac{\pi - \theta}{2}$	$\psi = \frac{\pi + \theta}{2}$

In all cases:

$$\boxed{\theta = |\pi - 2\psi|} \tag{3.10}$$

3.2.1 Computing $s(\theta, E)$: the Universal Method

$$E = \frac{m}{2} \dot{r}^2 + V(r) + \frac{\ell^2}{2mr^2}, \quad \dot{r} = \left[\frac{2}{m} \left(E - V(r) - \frac{\ell^2}{2mr^2} \right) \right]^{1/2}, \quad \dot{r} = \frac{dr}{dt} \Rightarrow dt = \frac{dr}{\dot{r}}. \tag{3.11}$$

$$\begin{aligned}
\ell = mr^2\dot{\theta} \Rightarrow \frac{d\theta}{dt} = \dot{\theta} = \frac{\ell}{mr^2} \Rightarrow d\theta = \dot{\theta} dt = \frac{\ell}{mr^2} dt = \frac{\ell}{mr^2} \frac{dr}{\left[\frac{2}{m} \left(E - V(r) - \frac{\ell^2}{2mr^2}\right)\right]^{1/2}} \\
= \frac{\ell}{m} \sqrt{\frac{m}{2E}} \frac{r^{-2} dr}{\left[1 - \frac{V(r)}{E} - \frac{\ell^2}{2mE} \frac{1}{r^2}\right]^{1/2}}. \quad (3.12) \\
= \frac{\ell}{\sqrt{2mE}} = s \left[1 - \frac{V(r)}{E} - \underbrace{\frac{\ell^2}{2mE}}_{=s^2} \frac{1}{r^2}\right]^{1/2}.
\end{aligned}$$

Integrating from the apsidal point out to infinity gives the apsidal angle ψ ;¹ the substitution $u = 1/r$ is then made.²

$$\begin{aligned}
\psi = \int_0^\psi d\psi = \int_{\theta_m}^\pi d\theta = \int_{r_m}^\infty \frac{s r^{-2} dr}{\left[1 - V(r) - s^2/r^2\right]^{1/2}}, \quad u = \frac{1}{r}, \quad du = r^{-2} dr, \quad u_m = \frac{1}{r_m} \\
\Rightarrow \theta(s) = \left| \pi - 2 \int_0^{u_m} \frac{s du}{\left[1 - \frac{V(1/u)}{E} - s^2 u^2\right]^{1/2}} \right| \quad (3.13)
\end{aligned}$$

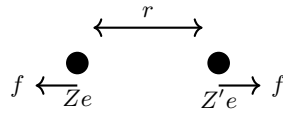
From here, compute

$$\left| \frac{ds}{d\theta} \right| = \frac{1}{\left| \frac{d\theta}{ds} \right|} \quad \text{and use it to get} \quad \sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right|. \quad (3.14)$$

Aside. Up to here, results are valid for any central potential $V(r)$.

3.3 Coulomb (Rutherford) Scattering

$$\text{Force} \quad f = \frac{ZZ'e^2}{r^2}, \quad (3.15)$$



$$f = -\frac{k}{r^2}, \quad V = -\frac{k}{r}, \quad k = -ZZ'e^2. \quad (3.16)$$

$E > 0$ hyperbolic $E = 0$ parabolic $E < 0$ elliptic Only for $V = -\frac{k}{r}$!!!
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¹The integrand of the r -integral below carries $V(r)$ rather than $V(r)/E$, while the boxed result on the same line correctly carries $V(1/u)/E$; the inconsistency is retained rather than normalized.

²Written as $u = 1/r$, $du = r^{-2} dr$; the missing sign on du is absorbed by the interchange of the limits of integration.

Scattering requires unbound trajectories, $E > 0$:

$$\frac{1}{r} = C[1 + \varepsilon \cos(\tilde{\theta} - \tilde{\theta}')] , \quad C = \frac{mk}{\ell^2} = -\frac{m}{\ell^2} ZZ'e^2. \quad (3.17)$$

$$\text{Eccentricity } \varepsilon = \left[1 + \frac{2E\ell^2}{mk^2}\right]^{1/2} = \left[1 + \frac{2E\ell^2}{m(ZZ'e^2)^2}\right]^{1/2} = \left[1 + \left(\frac{2sE}{ZZ'e^2}\right)^2\right]^{1/2} > 1, \quad (3.18)$$

$$\ell = smv_0 = s\sqrt{2mE} \rightarrow \frac{2\ell^2 E}{m} = \frac{s^2 2^2 \mathcal{M} E^2}{\mathcal{M}} = (2sE)^2. \quad (3.19)$$

Two situations:

- i) Repulsive case $ZZ' > 0 \Rightarrow k = -ZZ'e^2 < 0$;
- ii) Attractive case $ZZ' < 0 \Rightarrow k = -ZZ'e^2 > 0$ (also gravity, $k = Gm_1m_2$).

3.3.1 Repulsive Case: $k < 0$, $C = mk/\ell^2 < 0$

$$\begin{aligned} \frac{1}{r} = C[1 + \varepsilon \cos(\tilde{\theta} - \tilde{\theta}')] > 0 &\iff 1 + \varepsilon \cos(\tilde{\theta} - \tilde{\theta}') < 0 \\ &\iff \varepsilon \cos(\tilde{\theta} - \tilde{\theta}') < -1 \iff \cos(\tilde{\theta} - \tilde{\theta}') < -\frac{1}{\varepsilon}. \end{aligned} \quad (3.20)$$

We need the cosine to be negative and larger in magnitude than $1/\varepsilon$. In this case $\tilde{\theta} = \tilde{\theta}'$ is not a physical point, because

$$\cos(\tilde{\theta}' - \tilde{\theta}') = \cos(0) = 1 \implies \underbrace{\frac{1}{r} = C[1 + \varepsilon]}_{\text{unphysical}} < 0. \quad (3.21)$$

So $\tilde{\theta} = \tilde{\theta}'$ lies in the middle of the inaccessible region of the hyperbola, and $\tilde{\theta} = \tilde{\theta}' \pm \pi$ is the periapsis; we choose $\tilde{\theta}' = \pi$, giving

$$\cos(\tilde{\theta} - \tilde{\theta}') = \cos(\tilde{\theta} - \pi) = \cos \tilde{\theta} \underbrace{\cos \pi}_{-1} + \sin \tilde{\theta} \underbrace{\sin \pi}_0 = -\cos \tilde{\theta}. \quad (3.22)$$

$$\begin{aligned} \frac{1}{r} = \left(-\frac{mk}{\ell^2}\right) [-1 - \varepsilon \cos(\tilde{\theta} - \tilde{\theta}')] &= \frac{mZZ'e^2}{\ell^2} [\varepsilon \cos \tilde{\theta} - 1], & \text{maximum at } \tilde{\theta} = 0 \implies \\ \text{minimum } r : \frac{1}{r_m} &= \frac{mZZ'e^2}{\ell^2} [\varepsilon - 1]. \end{aligned} \quad (3.23)$$

So the angle at the apsidal point is $\tilde{\theta} = 0$. We now need

$$\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}(\frac{1}{r} \rightarrow 0), \quad \left[\text{repulsive case: } \theta = \pi - 2\psi \implies \psi = \frac{\pi - \theta}{2} \right] \quad (3.24)$$

$$0 = \varepsilon \cos \psi - 1 \iff \frac{1}{\varepsilon} = \cos \psi = \cos\left(\frac{\pi - \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} + \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = \sin \frac{\theta}{2}. \quad (3.25)$$

3.3.2 Attractive Case: $k > 0$, $C = km/\ell^2 > 0$

Take $\tilde{\theta}' = 0$:

$$\tilde{\theta} = \tilde{\theta}' = 0 \Rightarrow \frac{1}{r} = C[1 + \varepsilon \cos(0)] = C[1 + \varepsilon], \quad \begin{array}{l} \text{max at } \tilde{\theta} = 0 \\ \text{minimum } r : \frac{1}{r_m} = C[1 + \varepsilon]. \end{array} \quad (3.26)$$

The apsidal point is also at $\tilde{\theta} = 0$. We now need

$$\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}\left(\frac{1}{r} \rightarrow 0\right), \quad \left[\text{attractive case: } \theta = 2\psi - \pi \Rightarrow \psi = \frac{\pi + \theta}{2} \right] \quad (3.27)$$

$$0 = 1 + \varepsilon \cos \psi \iff -\frac{1}{\varepsilon} = \cos \psi = \cos\left(\frac{\pi + \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} - \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = -\sin \frac{\theta}{2}. \quad (3.28)$$

3.3.3 Combining Both Cases: the Rutherford Cross Section

Equations (3.25) and (3.28) give the same relation $1/\varepsilon = \sin(\theta/2)$ in both cases, so

$$\cot^2 \frac{\theta}{2} = \frac{1}{\tan^2 \frac{\theta}{2}} = \frac{\cos^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1 - \sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1}{\sin^2 \frac{\theta}{2}} - 1 = \varepsilon^2 - 1 = \left(\frac{2sE}{ZZ'e^2}\right)^2 \quad (3.29)$$

$$\implies \boxed{s = s(\theta, E) = \left| \frac{ZZ'e^2}{2E} \right| \cot \frac{\theta}{2}} \quad (3.30)$$

$$\frac{ds}{d\theta} = \left| \frac{ZZ'e^2}{2\varepsilon} \right| \frac{d}{d\theta} \left(\frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} \right) = \left| \frac{ZZ'e^2}{2\varepsilon} \right| \left(\frac{1}{2} \left(\frac{-\sin \frac{\theta}{2}}{\sin \frac{\theta}{2}} \right) - \frac{1}{2} \frac{\cos^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} \right) = - \left| \frac{ZZ'e^2}{4E} \right| \frac{1}{\sin^2 \frac{\theta}{2}},^3 \quad (3.31)$$

$$\sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right| = \frac{1}{2} \left(\frac{ZZ'e^2}{2E} \right)^2 \frac{1}{2 \sin \frac{\theta}{2} \cancel{\cos \frac{\theta}{2}}} \frac{\cancel{\cos \frac{\theta}{2}}}{\sin \frac{\theta}{2}} \frac{1}{\sin^2 \frac{\theta}{2}}. \quad (3.32)$$

$$\boxed{\sigma(\theta) = \frac{1}{4} \left(\frac{ZZ'e^2}{2E} \right)^2 \csc^4 \left(\frac{\theta}{2} \right)} \quad \begin{array}{l} \text{Rutherford scattering cross section} \\ \text{(for Coulomb forces)} \end{array} \quad (3.33)$$

Remarks:

- 1) The cross section is the same for the repulsive and attractive cases, since it is proportional to $(ZZ')^2$.
- 2) It diverges like E^{-2} at low energies.
- 3) It diverges like θ^{-4} (very strongly) for small angles θ .
- 4) This classical result is exactly the same as the result obtained in quantum mechanics.

³The first two forms carry ε in the denominator; the line switches to E (consistent with Eq. (3.30)) in the final simplification. The inconsistency is retained rather than normalized.

3.4 Total Cross Section

$$\sigma_T = \int \sigma(\Omega) d\Omega, \quad \text{in this case } \sigma(\theta) = \text{const} \frac{1}{\sin^4(\frac{\theta}{2})}. \quad (3.34)$$

$$d\Omega = 2\pi \sin \theta d\theta = 4\pi \sin \frac{\theta}{2} \cos \frac{\theta}{2} d\theta \quad (3.35)$$

$$\sigma_T = \frac{4\pi}{4} \left(\frac{ZZ'e^2}{2E} \right)^2 \int_0^\pi \frac{\sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\sin^4 \frac{\theta}{2}} d\theta, \quad \text{range: } 0 < \theta \leq \pi \quad (3.36)$$

(Fig. 3.3). Near $\theta = 0$ the integrand behaves as

$$\sim \int_0^\delta \frac{d\theta}{\left(\frac{\theta}{2}\right)^3} = 8 \int_0^\delta \frac{d\theta}{\theta^3} = 8 \left. \frac{\theta^{-2}}{(-2)} \right|_0^\delta = +\infty \quad \text{strongly divergent.} \quad (3.37)$$

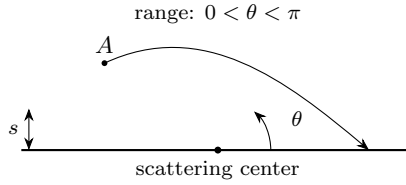


Figure 3.3: As θ ranges over $(0, \pi)$ the impact parameter sweeps its full accessible range about the scattering center.

NOTATION:			
<i>Goldstein</i>		<i>Others</i>	
$\sigma(\Omega)$	$\longleftrightarrow$	$\frac{d\sigma}{d\Omega}$	differential scattering cross section
σ_T	$\longleftrightarrow$	σ	total scattering cross section

Classical $\sigma_T = +\infty$ for any long-range potential.

Quantum $\sigma_T = +\infty$ for some long-range potentials $V(r)$ that do not decay fast enough as $r \rightarrow \infty$.

Caveat 1. In a dense conducting material the Coulomb potential is screened,

$$V = -\frac{k}{r} \longrightarrow V^{\text{screen}}(r) = -\frac{k}{r} e^{-r/\lambda}, \quad (3.38)$$

the screened Coulomb potential (“Yukawa”). In non-conducting materials $V^{\text{screen}}(r)$ may take a different form, but it still vanishes much faster than $1/r$.